

Model Exit Examination

Program: Bachelor of Science Degree in Health Informatics

Total mark __%

Time allowed __

Courses included:

- i. System Analysis and Design
- ii. Computer Maintenance and Troubleshooting
- iii. Fundamental of Programming
- iv. Object Oriented Programming
- v. Advanced Database Management and Administration
- vi. System and Network Configuration and Administration
- vii. Mobile Computing
- viii. Advanced Web Programming

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Based on given information select the appropriate answer for the following questions

Part I: System Analysis and Design

1.

Part II: Computer Maintenance and Troubleshooting

1. A company with a Windows-based network has recently grown in size and complexity. They are currently using a workgroup setup for authentication and authorization, but are now considering moving to a Windows Server domain. What would be the main advantage of this migration?
A) Increased security, as user accounts can be centrally managed and password policies enforced.
B) Increased network performance, as the domain controller will distribute traffic more efficiently.
C) Reduced costs, as the company will no longer need to purchase new servers.
D) Increased flexibility, as the company will be able to use multiple authentication methods.
2. Which of the following is a hierarchical database that stores information about objects on a network and makes this information available to users and administrators?
A) Windows Registry
B) Windows Active Directory
C) Domain Name System (DNS)
D) Lightweight Directory Access Protocol (LDAP)
3. You are setting up a new network for a small business and need to configure routing between different subnets. You want to ensure that traffic flows smoothly and securely between the subnets. Which of the following protocols is commonly used for this purpose?
A) Border Gateway Protocol (BGP)
B) Transmission Control Protocol (TCP)
C) User Datagram Protocol (UDP)
D) Internet Protocol (IP)
4. Which of the following types of proxies or gateways is commonly used for filtering and controlling internet access in a network?
A) Virtual Private Network (VPN)
B) Reverse Proxy
C) Content Filter Proxy
D) Load Balancer
5. Assume you are responsible for managing the performance of a database server that is critical to the company's operations. You want to ensure that the server can handle a high volume of

requests and has minimal downtime. Which of the following technologies will you implement to achieve this?

A) RAID 0

B) RAID 1

C) RAID 5

D) RAID 10

6. If you are a system administrator for a small business that has multiple remote sites and you want to be able to remotely manage the servers at these sites and perform routine maintenance tasks, which of the following tools will you use?

A) Remote Desktop Connection

B) Virtual Private Network (VPN)

C) Telnet

D) File Transfer Protocol (FTP)

7. You are responsible for managing a Linux mail server that is used by your organization. You want to configure the server to send and receive email securely. Which of the following protocols will you use to encrypt email transmissions?

A) SMTP

B) POP3

C) IMAP

D) TLS

8. Which of the following is NOT a security best practice for Linux systems?

A) Regularly updating software and security patches

B) Disabling unnecessary network services

C) Running processes with root privileges

D) Configuring a firewall

9. Which of the following services can be provided by a Linux server running SAMBA?

A) Web hosting

B) File and print sharing with Windows clients

C) Email hosting

D) Network time synchronization

10. Which of the following best describes the difference between a workgroup and a domain in a Windows network?

A) A workgroup is a collection of computers that share resources, while a domain is a collection of computers that are centrally managed.

B) A workgroup is a collection of computers that are centrally managed, while a domain is a collection of computers that share resources.

C) A workgroup and a domain both refer to the same thing in a Windows network.

D) A workgroup is a collection of computers that use a peer-to-peer networking model, while a domain is a collection of computers that use a client-server networking model.

11. Which of the following file systems is commonly used for managing storage on a Linux system?
- A) FAT32
 - B) NTFS
 - C) Ext4
 - D) HFS+
12. Which of the following resource monitoring and management tools is commonly used to measure network bandwidth usage?
- A) Performance Monitor
 - B) Resource Monitor
 - C) Wireshark
 - D) Netstat
13. Which of the following best describes the philosophy of system administration?
- A) Minimizing the number of users and devices on a network to increase security
 - B) Focusing on reactive rather than proactive maintenance to save time and resources
 - C) Prioritizing the needs of users and the organization to ensure reliable and efficient systems
 - D) Avoiding automation and relying solely on manual processes to reduce errors
14. Which of the following best defines a server domain?
- A) A group of workstations connected to a server on a peer-to-peer network
 - B) A group of servers connected to a central directory service
 - C) A decentralized network where each computer has equal administrative privileges
 - D) A virtual network that provides secure access to the internet
15. What is the purpose of a roaming profile in Windows user administration?
- A) To allow users to log in from any computer on the network
 - B) To synchronize user settings and data across multiple devices
 - C) To restrict user access to specific network resources
 - D) To enforce security policies and access controls for users
16. Which one of the following are the first operations that performed by system administrator?
- A) Startup
 - B) Shutdown
 - C) Check memory
 - D) A&B
17. The Windows boot procedure on a PC begins with the_____.
- A) BIOS
 - B) MBR
 - C) Shutdown button
 - D) NIC
18. Which one of the following is commercial software usually installed from a CD?
- A) Installation of software from binaries
 - B) Free and open software
 - C) Operating system with diagnostic

D) All

19. _____ is an additional copy of data that can be used for restore and recovery purposes

A) Mirroring

C) Backup

B) Logs file

D) Recycle bin

20. Loss of data can be happened due to _____

A) Accidental file deletion

C) Natural or manmade disaster

B) Disk failure

D) All

21. User account can be?

A) Created

C) Deleted

B) Maintained

D) All

22. What will be needs administrator as user management

A) Registration of new users

D) Ethical issues

B) Support services

E) All

C) Security

23. _____ is authenticates the user for local machine access only; access to resources on other computers is not supported?

A) User account

C) Local user account

B) Group account

D) network user account

24. from the following which user have full privileges to access computer resource

A) guest

C) standard

B) administrator

D) all

25. _____ is Policy rules required for guiding user behavior, and also for making system rules clear.

A) Group policy

C) Support policy

B) Account policy

D) All

26. _____ is the policy that makes users more self-sufficient through training and helping

A) Group policy

C) Support policy

B) Account policy

D) All

27. _____ is system a collection of tools and ruled for network monitoring and control

A) Network installation

C) Network design

B) Network configuration

D) Network management

28. _____ is abnormal condition that requires action to repair

- A) **Fault**
- B) security

- C) performance
- D) all

29. The mechanism by which Internet software translates names to addresses and vice versa is called

- A) TCP/IP
- B) DHCP

- C) **DNS**
- D) SNMP

30. ____ is an IP standard designed to reduce the complexity of administering IP address configurations?

- A) TCP/IP
- B) **DHCP**
- C) DNS
- D) SNM

Part III: Fundamental of Programming

1. High level languages are translated to machine language by either a _____ or a _____ respectively.
 - A. Assembler, compiler
 - B. assembler, interpreter
 - C. Compiler, interpreter**
 - D. Interpreter, compiler
 - E. compiler, assembler
2. Among the following, which one should be the early step in any system development process?
 - A. Identifying the users need
 - B. Implementing the solution
 - C. Designing the system to solve the problems
 - D. Testing the performance of the system
 - E. All
3. Suppose that n variable is declared to store the age of a person. How would you initialize the variable to 65?
 - A. float n = 65.0;
 - B. int n = 65;**
 - C. int n = "65";
 - D. int n = '65';

4. Based on the following code select the correct output

```
#include<iostream>
using namespace std;
int main ()
{
    int n,t,r,R=0;
    cout<<"Enter any number : ";
    cin>>n;
    t=n;
    while(t>0)
    {
        r=t%10;
        t=t/10;
        R=R*10+r;
    }
    cout<<"The O/P"<<n<<" is: "<<R;
    return 0;
}
```

- A. Enter any number: 4525, The O/P is: 4525
 - B. Enter any number: 4525, The O/P is: 4535
 - C. Enter any number: 4525, The O/P is: 5254
 - D. Enter any number: 4535, The O/P is: 4525
5. Consider the following code snippet:

```
#include<iostream>
using namespace std;
int main ()
{
    int year;
    cin>>year;
    if (year % 100==0)
        if (year %400 ==0)
            cout<<" leap,";
        else
            cout<<" not leap year,";
}
```

If we input1 = 300 and input2 = 2000 what will the output respectively

- A. leap, not leap,
 - B. leap, not leap year,
 - C. not leap year, leap,
 - D. not leap year leap
6. Let int k=40; and int m=5; what is the result of ++k +m, k++ +m, - -k +m, and k- --m
- A. respectively?
 - B. 46, 45, 45, 45
 - C. 45, 46, 44, 45
 - D. 46, 45, 44, 45
 - E. 46, 46, 44, 44
7. Find the output of the given c++ switch case program?

```
#include<iostream>
using namespace std;
int main()
{
    int a = 0;
    switch(a)
    {
        case 1:
            cout<<"Welcome";
            break;
        case 2:
            cout<<"Know Program";
            break;
        default:
            cout<<"Hello";
            break;
    }
    return 0; }
```

- A. Welcome
 - B. Know Program
 - C. Hello
 - D. Compile-time error
8. From the following levels of programming language, which one is different from the others?
- A. Low level Language
 - B. Binary Language
 - C. Assembly Language
 - D. Machine language
9. Which of the following line of program used to create the infinite loop using while loop?
- A. while (; ;)
 - B. while(true)
 - C. while(infinite)
 - D. while(false)

10. What is the output of the following Program?

```
int main() {  
    string course[2]={"C++", "Algebra", "Calculus", "Statistics"};  
    cout<<course[2]<<endl;  
}
```

- A. C++
 - B. Calculus
 - C. Algebra
 - D. Error
11. Which of the following is not a type of algorithm?
- A. Program**
 - B. Flowchart
 - C. Decision Table
 - D. Pseudocode
12. Which of the following statements is False?
- A. The flowchart should be clear, precise and easy to follow
 - B. The input, process, output, and display symbols should have only one flow line**
 - C. Flowcharts can have only one start point and one endpoint
 - D. The start and stop symbols should have only a single flow line
 - E. In a flowchart all the necessary inputs should be listed out in a logical order
13. Assembly language programs are written using
- A. Hex code
 - B. mnemonic**
 - C. ASCII code
 - D. None
14. ____ are different ways or styles in which a given program or programming language can be organized.
- A. Algorithm
 - B. Programming paradigm**
 - C. Pseudocode
 - D. Sequence of instructions
15. Here is a list of C++ variable names. choice which is valid expressions.

- A. `int num1 = 100,000;`
- B. `int num_1 = 100000;`**
- C. `int num 1 = 1000;`
- D. `int $num1 = 1000;`

16. All keyword in C++ is

- A. upper case
- B. Toggle
- C. Each word capitalization
- D. Lower case**
- E. All

17. Which one of the following is not primary data type in c++

- A. `int`
- B. `void`
- C. `array`**
- D. `float`

18. Indicate whether the following statements are true or false

Statement 1: `enum` is user-defined data type consisting of a set of named constants called enumerator

Statement 2: logical operators are used to test only two condition and makes decisions

Statement 3: bitwise operators are used to perform operations bit by bit.

- A. Statement 1 and statement 2 are true while statement 3 is false
- B. Statement 2 and statement 3 are true while statement 1 is false
- C. Statement 1 and statement 3 are true while statement 2 is false**
- D. All statements are true

19. Which one of the following is used as an extraction operator in c++

- A. `(>>)`**
- B. `(<<)`
- C. `(&&)`
- D. `(&->)`
- E. `(&*)`

20. Which operators are used to test the relationship between two variables or between a variable and constant.

- A. Arithmetic operators
- B. Logical operators
- C. Relational operators**
- D. Assignment operators

21. If an expression contains an addition operator, a less than operator, a subtraction and a multiplication operator, which operator has a higher priority when evaluating the expression?
- A. Addition operator
 - B. Multiplication operator**
 - C. Subtraction operator
 - D. Less than operator
22. Which one of the following is relational operator?
- A. $x=y$
 - B. $x\%y$
 - C. $z!=y$**
 - D. x/y
23. Which of the following is alter the flow of program execution?
- A. while**
 - B. count
 - C. #define
 - D. Function
24. If x contains the value 3 before the following instruction is executed, what is the value of x after the instruction: $x*=5$; is it executed?
- A. Compiler error
 - B. Variable error
 - C. 15**
 - D. 5
 - E. 25
25. Which of the following is an illegal variable name?
- A. `_myName`
 - B. `xName`
 - C. `P2P`
 - D. `2x_value`**
26. Which numeric data type has the largest range
- A. int
 - B. char
 - C. float
 - D. double**
27. Consider the following code fragment, then answer the question: how many times will the count statement execute:
- ```
for(int i= 0; i<=5; i++){
 {
 cout<< i;
```

}

- A. 5 times
  - B. 4 times
  - C. 6 times**
  - D. compiler error
  - E. 1 times
28. What are the three basic control structures used in programming?
- A. int, double, string
  - B. while, do...while, and for
  - C. sequence, decision, and repetition**
  - D. input, output, and calculation
29. A loop exit condition must
- A. Be the last instruction in the body of the loop
  - B. Evaluate to true or false**
  - C. Be the first instruction in the body of the loop
  - D. Not use compound conditions
30. What is the output the following c++ code?

```
#include<iostream>

using namespace std;

int main ()
{
 int i=1, sum=0;
 while(i<=10)
 {
 sum=sum+1;
 i++;
 }
 cout<<sum;
}
```

- A. 1
- B. 10
- C. 1,2,3,4,5,6,7,8,9,10,
- D. 55**

- E. 1,2,3,4,5,6,7,8,9
31. The keyword 'break' cannot be simply used within:
- A. do-while
  - B. if-else**
  - C. for
  - D. While
32. Which keyword is used to come out a loop only for that iteration
- A. break
  - B. goto
  - C. continue**
  - D. return
  - E. None
33. What is the output of this code?
- ```
#include<iostream>

using namespace std;

void main ()

{

    int k=0;

    for(k)

        cout<<"hi";

}
```
- A. hi
 - B. Compile time error**
 - C. null
 - D. nothing
 - E. 0
34. Which of the following statements are false?
- A. bool can have two values and can be used to express logical expressions
 - B. bool cannot be used as the type of the result of the function**
 - C. bool can be converted into integers implicitly
 - D. bool value can be used in arithmetic expressions
 - E. None

Part IV: Object Oriented Programming

1. What will be the output of the following Java code snippet?

```
class Student
{
    int marks;
    Student () { }
    student (int x)
    {
        marks=x;
    }
    public static void main (String [] args) {
        Student s1 = new Student (100);
        Student s2 = new Student ();
    } }
```

- A. Program will give compile time error
B. Object s3, syntax error
C. Only objects s1 and s2 will be created
D. Program runs and all objects are created
2. What is encapsulation in Object oriented programming?
- A. It is a way of combining various data members and member functions that operate on
B. those data members into a single unit
C. It is a way of combining various data members and member functions into a single unit
D. that can operate on any data
E. It is a way of combining various data members into a single unit
F. It is a way of combining various member functions into a single unit
3. What is the extension of the following java code files?

```

class Complement {
public static void main (String args[])
{
    int x = 8;
    System.out.print(~x);
}

```

- A. 10
 - B. -8
 - C. 7
 - D. -9
4. What is the output of the following java code snippet?

```

class Parent {
    int id =1;
    void print (){
        System.out.println(id);
    } }
class Child extends Parent {
    int id =2;
}
public class Finalexam {
    public static void main (String [] args) {
        Child c = new Child ();
        c.print(); }}

```

- A. 2
 - B. 1
 - C. 0
 - D. Nothing
5. What is the output of the following java code snippet?

```

class A {
    public A () {
        System.out.println("Good Luck!");
    } }
class B extends A {
    public B () {
        System.out.println("Happy Java coding!");
    } }
public class C {
    public static void main (String [] args) {
        B b = new B ();
    }
}

```

- A. Good Luck!
 - B. Good Luck!
 - Happy Java coding!
 - C. Happy Java coding!
 - D. Happy Java coding!
 - Good Luck!
6. From the following file open mode which one is used to deletes all previous content in the file
 - A. ios::nocreate
 - B. ios::replace
 - C. ios::trunc
 - D. ios::noreplace
 7. Which is correct syntax?
 - A. myfile:open ("example.txt", ios::out);
 - B. myfile.open ("example.txt", ios::out);
 - C. myfile::open ("example.tx", ios::out);
 - D. myfile.open ("example.txt", ios:out);
 8. Which is not a feature of Object Oriented Programming in general definitions?
 - A. Efficient Code
 - B. Code reusability
 - C. Modularity
 - D. Duplicate/Redundant data
 9. Which feature of Object Oriented Programming indicates code reusability?
 - A. Abstraction
 - B. Polymorphism
 - C. Encapsulation
 - D. Inheritance
 10. Which of the following is used to create a stream that performs both input and output operations?
 - A. Ofstream
 - B. Ifstream
 - C. Iostream
 - D. Fstream
 11. Which of the following refers to characteristics of an array?
 - A. An array is a set of distinct data items
 - B. An array can hold different types of datatypes
 - C. An array is a set of similar data items**
 - D. None of the above
 12. Among the following which one is correctly an array declaration?

A. int array [10];	C. array [10];
B. int array;	D. array array [10];
 13. What is the index of an array with 5 elements?

1. 9
 2. 5
 3. 0
 4. **4**
14. Let we have a pointer variable called j then, which one of the following is odd
1. &j means the address of j
 2. ***j means the content pointed by &j**
 3. j means the content of j
 4. None
15. A string is terminated by a
1. Semicolon
 2. **Null character**
 3. Colon
 4. All
16. _____ is user-defined type which group variables as well as meaning a variable based on a user-defined type.
- A. Function
 - B. **Structure**
 - C. Array
 - D. Method
17. Which one of the following is not true about structure?
- A. Data members are also known as member variables
 - B. The storage unit is not reserved until you create a variable with it
 - C. **We can initialize member variables in the structure definition**
 - D. Member variables are distinguished by the structure variable
18. Which one of the following is false about ios::app file mode
- A. All the data you write, is put at the end of the file
 - B. **It calls ios::in**
 - C. It calls ios::out
 - D. Append mode
19. The data elements in the structure are also known as _____
- A. objects
 - B. **members**
 - C. data
 - D. data types
20. What will be used when terminating a structure?
- A. :
 - B. }
 - C. ;
 - D. :

21. What will happen when the structure is declared?
- A. it will not allocate any memory**
 - B. it will allocate the memory
 - C. it will be declared and initialized
 - D. it will be declared
22. Which of these is the correct statement about eof() ?
- A. Returns true if a file open for reading has reached the next character.
 - B. Returns true if a file open for reading has reached the next word.
 - C. Returns true if a file open for reading has reached the end.**
 - D. Returns true if a file open for reading has reached the middle.
23. Which of these cannot be used for a variable name ?

- | | | | |
|----|-----------------------|---|----------------|
| A) | Identifier | & | keyword |
| B) | | | identifier |
| C) | | | keyword |
| D) | none of the mentioned | | |

24. what the following array snapshot code output?

```
#include<iostream>

using namespace std;

int main()
{
    int array[10];

    for(int i=0; i<10; i++)
    {
        cout<<i<<array[i]<<endl;
        break;
    }

    return 0;
}
```

- A. 0
- B. 00**
- C. 01
- D. 10

25. What will be output of the following code when executed

```
#include <iostream>
using namespace std;
int main(){
    int i = 50, j = 75;
    int *p1, * p2 ;
    p1 = &i ;
    p2 = &j;
    cout << *p1<<endl;
    p1 = p2 ;
    *p2 =0;
    cout <<*p1;
    return 0;
}
```

- A. 75 and 0 respectively

- B. 0 and 50 respectively
- C. 50 and 75 respectively
- D. 50 and 0 respectively**
- E. 75 and 50 respectively

Part V: Advanced Database Management and Administration

1. Which of the following is **incorrect** statement?
 - A. Database is random assortment of data**
 - B. Database represents some aspect of the real world
 - C. Database is designed, built, and populated with data for a specific purpose.
 - D. All of the Above
2. Which one of the following commands is used for removing (or deleting) a relation from the SQL database?
 - A. Delete
 - B. Drop**
 - C. Remove
 - D. All of the above
3. Which of the following refers to collection of the information stored in a database at a specific time?
 - A. Independence
 - B. Instance of the database**
 - C. Schema
 - D. Data domain
4. Which of the following is not considered to be a basic element of an enterprise-class database system?
 - A. Users
 - B. Database applications
 - C. DBMS
 - D. COBOL program**
5. All of the following are properties of relational database except _____.
 - A. Each tuple in a relation must be unique.

- B. Each cell of the relation contains one value
 - C. The order of the attributes and tuples is significant
 - D. Each attribute has distinct names
6. The following are functions of a DBMS except _____.
- A. Creating and processing forms
 - B. Creating databases
 - C. Processing data
 - D. Administrating database
7. The DBMS acts as an interface between what two components of an enterprise-class database system?
- A. Database application and the database
 - B. Data and the database
 - C. The user and the database application
 - D. Database application and SQL
8. Which of the following is known as a set of entities of the same type that share same properties, or attributes?
- A. Relation set
 - B. Tuples
 - C. Entity set
 - D. Entity Relation model
9. Find the cities name with the condition and temperature from table 'whether' where condition = sunny or cloudy but temperature >= 60.
- A. SELECT city, temperature, condition FROM weather WHERE condition = 'cloudy' AND condition = 'sunny' OR temperature >= 60
 - B. SELECT city, temperature, condition FROM weather WHERE condition = 'cloudy' OR condition = 'sunny' OR temperature >= 60
 - C. SELECT city, temperature, condition FROM weather WHERE condition = 'sunny' OR condition = 'cloudy' AND temperature >= 60
 - D. SELECT city, temperature, condition FROM weather WHERE condition = 'sunny' AND condition = 'cloudy' AND temperature >= 60

10. The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation, according to _____ integrity constraint.
- A. Primary
 - B. Referential
 - C. Referencing
 - D. Specific
11. The traditional storage of data organized by the customer, stored in separate folders in filing cabinets is an example of _____ type of 'database' management system.
- A. Object-oriented database management system
 - B. Relational database management system
 - C. Network database management system
 - D. Hierarchical database management system
12. Which of the following refers to the level of data abstraction that describes exactly how the data actually stored?
- A. Conceptual Level
 - B. Physical Level
 - C. File Level
 - D. Logical Level
13. If attribute **A** determines both attributes **B and C**, then it is also true that:
- A. $A \rightarrow B$.
 - B. $B \rightarrow A$.
 - C. $C \rightarrow A$.
 - D. $(B, C) \rightarrow A$.
14. The attribute name could be structured as an attribute consisting of first name, middle initial, and last name. This type of attribute is called
- A. Multivalued Attribute
 - B. Derived Attribute
 - C. Composite Attribute
 - D. Atomic Attribute

15. One of the following is used to create a dynamic subset of rows/columns from one or more tables
- A. Create Index
 - B. Create View**
 - C. Create table as
 - D. Create schema
16. Which of the following is the correct order of a SQL statement?
- A. SELECT, GROUP BY, WHERE, HAVING
 - B. SELECT, WHERE, GROUP BY, HAVING**
 - C. SELECT, HAVING, WHERE, GROUP BY
 - D. SELECT, WHERE, HAVING, GROUP BY
17. Suppose a bank employee transfers 1000 birr from A's account to B's account and the employee applied rollback command at the end of the transaction. From this description what will be the new balance of account B?
- A. $\text{New_balance} = \text{Old_balance} - 1000$
 - B. $\text{New_balance} = \text{Old_balance} + 1000$
 - C. $\text{New_balance} = \text{Old_balance}$**
 - D. $\text{New_balance} = \text{Unknown}$
18. The database should be existed enough to hold all its latest updates even if the system fails or restarts. This property of a database transaction is _____.
- A. Atomicity
 - B. Consistency
 - C. Isolation
 - D. Durability**
19. Which command is used for removing all its data, but not the structure of the table from the database:
- A. Delete command
 - B. Drop table command
 - C. Truncate table command**
 - D. All of the above
20. Which of the following Statement is Incorrect?
- A. Distributed database architecture provides greater efficiency and better performance
 - B. Distributed database system provides local autonomy

C. The data redundancy is minimal in centralized database.

D. The data redundancy is minimal in distributed database

21. Which one of the following commands is used to modify a column inside a table?

A. Drop

B. Update

C. Alter

D. Set

22. The fourth normalization form is designed to cope with_____.

A. Transitive dependency

B. Joint Dependency

C. Multivalued dependency

D. Partial dependency

23. The preface that converts the DML statements to normal procedure calls in the host language, is known as_____.

A. DML pre-executor

B. DML executor

C. DML compiler

D. DML precompile

24. _____ is a program that performs some common action on database data and also stored in the database.

A. Stored Procedure

B. Trigger

C. Stored Function

D. None of the above

25. The file organization that provides very fast access to any arbitrary record of a file is__.

A. Ordered file

B. Unordered file

C. B-tree

D. Hashed file

26. Which of the following is used in the application programs to request data from the database management system?

A. Data Manipulation language

B. Data Definition Language

C. Data Control Language

D. All of the above

27. If a multivalued dependency holds and is not implied by the corresponding functional dependency, it usually arises from one of the following sources.

A. many-to-many relationship set

B. A multivalued attribute of an entity set

C. A one-to-many relationship set

D. Both A many-to-many relationship set and multivalued attribute of an entity set

28. The refinement from an initial entity set into successive levels of entity subgroupings represents a _____ design process in which distinctions are made explicit.

A. Hierarchy

B. Bottom-up

C. Top-down

D. Radica

29. The designer can review the schema to ensure it meets all the functional requirements, at the stage

A. Execution design

B. Conceptual design

C. Controlling design

D. Initial Planning

30. Consider the employee work-team example, and assume that certain employees participate in more than one work team. A given employee may therefore appear in more than one of the team entity sets that are lower level entity sets of employee. Thus, the generalization is _____.

A. Overlapping

B. Disjointness

C. Uniqueness

D. Relational

31. The query given below give an error. Which one of the following is the reason for the error?

```
SELECT ID, employee name, dept, salary * 1.1
WHERE dept = 'IT';
```

- A. The query have no **From** clause to indicate the table to select from
 - B. The column 'employee name' should be quoted
 - C. The 'salary * 1.1' is not applied correctly
 - D. A and B
 - E. All of the above
32. Which is the correct order for a proper SQL query?
- A. **SELECT, FROM, WHERE, GROUP BY, ORDER BY, HAVING**
 - B. **SELECT, FROM, WHERE, HAVING, ORDER BY, GROUP BY**
 - C. **SELECT, FROM, WHERE, HAVING, GROUP BY, ORDER BY**
 - D. **SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY**

Refer to the following 'Employee' table to answer the following questions.

Employee_id	Name	Salary	Department
1001	Abebe	6000	IT
1009	Almaz	4500	Sales
1018	Belay	7000	IT
1002	Jemal	5500	IT
1006	Yosef	6000	Sales

33. If we want to get the department name and the highest salary in each department, which is the correct query?

- A. **SELECT TOP** department, salary **FROM** Employee
WHERE Department='IT' AND Department='Sales';
- B. **SELECT** department, **MAX**(salary) **FROM** Employee
GROUP BY department;

C. **SELECT** department, salary **FROM** Employee
GROUP BY department;

D. **SELECT TOP** department, **MAX**(salary) **FROM** Employee
WHERE Department='IT' AND Department='Sales';

Answer: B

34. If we want to get the name of the employee from IT department in highest to lowest salary order, what is the correct query?

A. **SELECT * FROM** Employee
WHERE Department='IT' **ORDER BY** salary **ASC**;

B. **SELECT** name, salary **FROM** Employee
WHERE Department='IT' **ORDER BY** salary **DESC**;

C. **SELECT** name **FROM** Employee
WHERE Department='IT' **ORDER BY** salary **ASC**;

D. **SELECT** name **FROM** Employee
WHERE Department='IT' **ORDER BY** Salary **DESC**;

Answer: D

Refer to the following '*Employee_Department*' table by associating it to the previous '*Employee*' table to answer the next question.

dep_id	Name	Location
IT	Information Technology	building_1
sales	Sales department	building_2
mgmt	Management	building_3

35. What is the correct query if we want to find the name of the employee working in building_2 and the name of their respective department? Note that, '*department*' is the foreign key of employee table referencing employee_department table 'dep_id' column.

A. **SELECT** emp.name, emp.department **FROM** Employee emp
INNER JOIN Employee_Department' dep
ON emp.department = dep.dep_id
WHERE location='building_2';

B. **SELECT** name, department **FROM** Employee

UNION

SELECT FROM Employee_Department'
WHERE location='building_2';

C. **SELECT** emp.name, emp.department **FROM** Employee emp
INNER JOIN Employee_Department'
WHERE location='building_2';

D. **SELECT** emp.name, emp.department
WHERE location='building_2';
FROM Employee emp
INNER JOIN Employee_Department' dep
ON emp.department = dep.dep_id

Answer: A

36. Which of the following is the correct syntax for defining a parameter in the stored procedure?
- A. MODE param_name param_type(param_size)
 - B. (BEFORE | AFTER)_tableName_(INSERT| UPDATE | DELETE)
 - C. Tablename_(BEFORE | AFTER)_(INSERT| UPDATE | DELETE)
 - D. MODE param_type param_name (param_size)
37. Which of the following is Not the correct statement regarding good Concurrency Protocol
- A. It allows the parallel execution of transactions to achieve maximum concurrency.
 - B. Its storage mechanisms and computational methods should be modest to minimize overhead
 - C. It must enforce some constraints on the structure of atomic actions of transactions.
 - D. Must be resilient to site and communication failures.
 - E. All of the above
 - F. None of the above
38. A transaction that is inserting a new tuple in to the database is given.....?
- A. Shared lock
 - B. Mutual lock
 - C. Exclusive lock
 - D. Binary lock
39. Which one of the following contains only application layer languages or tools?

- A. C++, HTML5, Python, C#
 - B. MYSQL, Oracle, Python, CSS
 - C. C#, Java script, MongoDB, C++
 - D. C++, Java, Python, C#
40. One of the following is Not true about triggers
- A. Triggers are invoked and executed invisible from the client applications
 - B. Triggers can provide an extended validation and they can replace all the validations.
 - C. Triggers are very useful to audit the changes of data in tables
 - D. Triggers can catch errors in business logic in the database layer.
41. _____ is the locking protocols determine the order between every pair of conflicting transactions at execution time by the first lock that both members of the pair request that involves incompatible modes
- A. Pre-claiming Locking Protocol
 - B. Simplistic Locking Protocol
 - C. Timestamp-based Protocol
 - D. Shared/exclusive Protocol
42. Which of the following is incorrect statement?
- A. Recovery from failure is more complex in distributed database systems than in centralized systems.
 - B. In Parallel database nodes can only work together to handle global transactions.
 - C. Client/Server environment facilitates in more productive work by the users and making better use of existing data.
 - D. In Distributed database sites can work dependently to handle local transactions and work together to handle global transactions.
43. _____ is a hardware malfunction, or a bug in the database software or the operating system, that causes the loss of the content of volatile storage, and brings transaction processing to a halt.
- A. System Crash
 - B. Transaction failure
 - C. Disk failure

- D. Fail-stop assumption
44. Suppose a bank employee transfers 1000 birr from A's account to B's account and the employee applied commit command at the end of the transaction. From this description what will be the new balance of account B?
- A. $\text{New_balance} = \text{Old_balance} - 1000$
 - B. $\text{New_balance} = \text{Old_balance} + 1000$
 - C. $\text{New_balance} = \text{Old_balance}$
 - D. $\text{New_balance} = \text{Unknown}$ E. None of the above
45. The database should be existed enough to hold all its latest updates even if the system fails or restarts. This property of a database transaction is _____.
- A. Atomicity
 - B. Consistency
 - C. Isolation
 - D. Durability
46. _____ is a locking mechanism that acquired to perform a write operation on a data item.
- A. Shared Lock
 - B. Exclusive Lock
 - C. Two Phase Lock
 - D. Binary Lock
47. What is the function of using “DELIMITER //” in writing stored procedures?
- A. To change the standard delimiter semicolon (;) to another.
 - B. To change semicolon (;) to double-slashes //
 - C. To start the body of the stored procedure
 - D. To query data from the a specified table
48. Which of the following is Not true about checkpoint concurrent transaction recovery?
- A. It is a mechanism where all the previous logs are removed from the system and stored permanently in a storage disk.
 - B. It declares a point before which the DBMS was in consistent state, and all the transactions were committed.
 - C. It Keeps and maintain logs in real time and in real environment

- D. It reads the logs backwards from the end
 - E. None of the above
49. A transaction in which either all of the database actions occur or none of them do is _____:
- A. Atomic
 - B. Consistent
 - C. Isolated
 - D. Durable
50. which of the following is Incorrect Statement?
- A. Distributed database architecture provides greater efficiency and better performance
 - B. Distributed database system provides local autonomy
 - C. The data redundancy is minimal in the centralized database.
 - D. The data redundancy is minimal in the Distributed database.
51. which procedure parameter enables the caller to pass in a value and get back a value?
- A. IN
 - B. OUT
 - C. INOUT
 - D. GET_IN_OUT
52. Which of the following locks the item from change but not from read?
- A. Implicit lock
 - B. Explicit lock
 - C. Exclusive lock
 - D. Shared lock

Part VI: System and Network Configuration and Administration

1. Everyone in the company has a computer, but the business manager has the only printer. These computers are not connected by any form of networking. When agents need to print document, they must first copy the file to a flash disk, then carry it to the business manager's computer, where they are finally able to print it. Similarly, when staff members want to share data, the only means available is to copy the data on one computer to a flash disk and insert the disk in another computer. Recently, problems have arisen. The business manager is spending too much time printing other people's documents; and it is frequently unclear which copy of a given document is the current and authoritative version. As an IT expert,

your task is to design a network for this company. Which network topology would be the most appropriate in this situation?

- A. Bus Topology
 - B. Ring Topology
 - C. Star Topology**
 - D. Mesh Topology
2. Which one of the following statement is false about identifying the class in binary notation?
- A. If the first bit of the first byte is 0 then it is Class A address
 - B. If the first three bits of the first byte are 111 then it is Class B address
 - C. If the first three bits of the first byte are 110 then it is Class C address
 - D. If the first four bits of the first byte is 1110 then it is Class D address
3. From the following alternatives, which one is the correct statement regarding the IP address?
- i. We can use IP address with network bits all 0's
 - ii. We can use IP address with network bits all 1's
 - iii. We cannot use IP address with network bits all 1's
 - iv. We cannot use IP address with network bits all 0's
 - v. We can use IP address with host bits all 1's
 - vi. We can use IP address with host bits all 0's
- A. Only i and ii are correct
 - B. Both ii, iii, and v are correct
 - C. Both ii, iii, iv, v and vi are correct
 - D. Only iii and iv are correct
4. An organization is assigned the network ID 114.0.0.0 by Inter- NIC. The organization currently has 5 subnets with about 100,000 hosts per subnet. The management wants to divide the subnets into 25 new subnets to make each subnet more manageable. How many bits are used for the custom subnet mask?
- A. 4
 - B. 5
 - C. 6
 - D. 7
5. Using the following address and subnet mask 195.106.14.0/24, what is the total number of networks and the total number of host per network?
- A. 1 network with 254 hosts.
 - B. 2 networks with 128 hosts.
 - C. 4 networks with 64 hosts.
 - D. 6 networks with 30 hosts.
6. Suppose that you hired in the XYZ Company as the network administrator. For the purpose of security, you want to implement the VLAN in this company's network system. In which level of the OSI model you are dealing with?
- A. Physical link layer
 - B. Data link layer
 - C. Network Layer

- D. Transport layer
- 7. From the following alternatives, all concepts are true except one:
 - A. Router is implemented at the network layer of OSI model
 - B. Hub is implemented at the physical layer of OSI model
 - C. Firewall is implemented at the session layer of OSI model
 - D. Switch is implemented at the data link layer of the OSI model
- 8. Which of the following statement is correct regarding the IP address classes?
 - i. Class C IP address ranges from 192 to 224
 - ii. Class A IP address ranges from 1 to 127
 - iii. Class D IP address is reserved for future use (research)
 - iv. Class E IP address ranges from 240 to 255
 - v. Class D IP address is reserved for Multicasting purpose
 - A. Only i and iv is correct
 - B. Both iii and v is correct
 - C. Both ii and iv is correct
 - D. ii, iv, and v are correct
- 9. Given an IP address of 131.107.2.160 and a subnet mask of 255.255.255.192, to which subnet does the host belong?
 - A. A. 131.107.2.64
 - B. B. 131.107.2.96
 - C. 131.107.2.128
 - D. 131.107.2.192

Part VII: Mobile Computing

1. _____ is involves the process of transmitting information through invisible waves in the air.

A. Mobile device	C. Wireless communication
B. Android	D. All
2. What are the physical representation of data?

A. Bandwidth	D. Data rate
B. Signals	E. None
C. phase	
3. _____ is electronics in the satellite that receive a signal on one frequency and amplify the signal and transmit it on another frequency.

A. Footprint	D. Spectrum
B. Doppler affect	E. None
C. Transponder	
4. Signal propagation can be influenced by:

A. Large obstacle	D. None
B. Distance	E. All
C. Type of medium	
5. Wireless communication system consists of :

A. Transmitters	D. All
B. Antennas	E. None
C. Receivers	
6. A type of antenna which send and receive radio signals from all direction is called

A. Transmitter	C. Omnidirectional
B. Transceiver	D. Directional
7. Which one of the following is odd?

A. Ground wave	
B. Sky wave	
C. Isotropic radiator	
D. Line –of-sight	

8. Which one of the following is types of wireless communication services?
- A. broadcast
 - B. paging
 - C. cellular telephony
 - D. trunking radio
 - E. **all**
9. Which one is different
- A. Data rate
 - B. Cordless telephony
 - C. **Mobility**
 - D. Use of spectrum
10. Which one is an application area of wireless communication
- A. Ease of setup
 - B. Less expensive
 - C. Increase revenue
 - D. **Business**
11. From the following which one is not the voice communication services of GSM
- A. Basic telephony
 - B. Emergency number
 - C. **Fax**
 - D. Voice mailbox
12. A transmission media that send and receive information through solid medium is;
- A. Fiber optic
 - B. Unguided
 - C. **Guided**
 - D. Wireless
13. From the following cellular hierarchy, which is covers few tens of meters
- A. Microcells
 - B. Bluetooth
 - C. Wifi
 - D. **Picocells**
14. The ability for a cellular customer to automatically make and receive voice calls, send and receive data, or access other services when travelling outside the geographical coverage area of the home network is.
- A. MSC
 - B. HLR
 - C. **Roaming**
 - D. PSTN
15. The transmission from the earth to satellite is
- A. Downlink
 - B. Upload
 - C. **Uplink**
 - D. Earth station
16. _____ is part of antenna that receive a signal on one frequency and amplify the signal and transmit it on another frequency.
- A. Antenna
 - B. **Transceiver**
 - C. Transponder
 - D. Signal

17. Data may be difficult to be identified (corrupted or mis-interpreted) due to;

- A. **Noises**
- B. Interferences
- C. Attenuation
- D. All
- E. None

18. A signal propagation occurs when signal encounters large surface is

- A. Diffraction
- B. Scattering
- C. **Reflection**
- D. Attenuation

19. Which one of the following is type of multiplexing?

- A. FDM
- B. TDM
- C. SDM
- D. **All**
- E. None

20. Which is not components of GSM operation subsystem

- A. Authentication Center
- B. Equipment Identity Register
- C. Operation and Maintenance Center
- D. **Location registration center**

21. ____ is types of mobile services that GSM offers
- A. **Voice communication**
 - B. Large file sharing
 - C. Multiple time slot
 - D. All
22. Which one of the following is not types satellite?
- A. GEO
 - B. MEO
 - C. LEO
 - D. **None**
23. Which one of the following is role of mobile services switching center?
- A. Hand over
 - B. Management of network resource
 - C. Accounting and billing information
 - D. **All**
24. ____ is the time required for a satellite to make a complete trip around the earth.
- A. Orbit
 - B. Transponder
 - C. Transceiver
 - D. Period
 - E. **None**
25. ____ is the area that the signal from a satellite is aimed at (the signal power is maximum)
- A. Orbit
 - B. Footprint
 - C. **Earth station**
 - D. Uplink

26. Suppose that you are calling with somebody over the telephone call. At the time of call, both of you can talk at the same time without connection interruption. Which type of data transmission mode we classify this scenario?
- A. Full duplex**
 - B. Simplex
 - C. Half duplex
 - D. A and C

Part VIII: Advanced Web Programming

1. ____ is a document, typically written in HTML that is almost always accessible via HTTP.
 - A. Hyperlink
 - B. Web page**
 - C. Hypertext
 - D. www
2. What is the correct way to write a JavaScript array?
 - A. var colors = 1 = ("red"), 2 = ("green"), 3 = ("blue")
 - B. var colors = "red", "green", "blue"
 - C. var colors = ["red", "green", "blue"]**
 - D. var colors = (1:"red", 2:"green", 3:"blue")
3. How do you write a do-while loop in JavaScript?
 - A. do { i++; } while (i <= 5)**
 - B. do { i++; } while i <= 5
 - C. do i++; while (i <= 5)
 - D. do i++; while i <= 5
4. What is the correct HTML element for creating a hyperlink?
 - A. link text**
 - B. <link>link text</link>
 - C. <hyperlink>link text</hyperlink>
 - D. <a>link text
5. _____ is a protocol that supports communication between web browser and web server.
 - A. FTP
 - B. HTTP**
 - C. HTML
 - D. TCP/IP
 - E. All
6. How do you write a switch statement in JavaScript?
 - A. switch (i) {case 0: x = "zero"; break; case 1: x = "one"; break;}**
 - B. switch i {case 0: x = "zero"; break; case 1: x = "one"; break;}
 - C. switch (i) {case 0 x = "zero"; break; case 1 x = "one"; break;}
 - D. switch i {case 0 x = "zero"; break; case 1 x = "one"; break;}
7. ____ is used to divide part of page in to columns keeping layout like newspapers.
 - A. Span
 - B. div**

- C. class
 - D. id
8. ___ is used for validations and functionality for user events
- A. Server side scripting
 - B. Client side scripting
 - C. Web server
 - D. Browsers
9. ___ is provide an interface to the user and limit access to proprietary data and help keep control of the script source code.
- A. Server side scripting
 - B. Client side scripting
 - C. Web server
 - D. Browsers
10. Which of the following is properties of PHP programming language?
- A. Compatible with almost all servers
 - B. Run on most platforms
 - C. Supports wide range of databases
 - D. All
11. From the following which one is correct PHP syntax

- A. <?....?>
 - B. <? php?>
 - C. <script language = "PHP"> </script>
 - D. All
12. What is the output of the following code?
- ```

<? php
 $num1 = 5;
 $num2 = 2;
 $num1 -= $num2;
 echo $ num1;
?>

```
- A. 2
  - B. 5
  - C. 3
  - D. 4
13. Which one of the following is used to checks for both value and data type?
- A. ==
  - B. ===
  - C. ===
  - D. >=
14. Which one of the following is important to organize information on a single page or across website?
- A. Location
  - B. Category
  - C. Time
  - D. Continuum
  - E. All
15. \_\_\_\_ is defines the interactions between applications, middleware systems and databases
- A. Server –side scripting
  - B. Client-side scripting
  - C. Web application architecture
  - D. N-tier Architecture
16. From the following which is static or dynamic generated content rendered by the browser

- A. Tier
- B. Logical layer
- C. Presentation layer
- D. Data layer

17. When applying inline style, style rules are added to the HTML using

- A. <style>tag in the head section
- B. Style in the html tag
- C. <link> tag in the head section
- D. A and C

18. \_\_\_\_\_ is used to divide part of page into columns keeping layout like newspapers

- A. span
- B. div
- C. class
- D. id

19. which one of the following is not true about array in PHP?

- A. array name can be any variable based on variable naming rules
- B. \$var\_name[ ] used to add element to array
- C. unset (\$var\_name[ ]) used to remove element
- D. none

20. let we have the following array example;

```
<?php
$Bodymassindex=array ("weight "=>"65kg","Height"=>"1.8cm","BMI"=>"20.3");
echo"your weight is". $ Bodymassindex [weight]."
";
echo"your height is". $ Bodymassindex [height]. " and
";
echo"your BMI is". $ Bodymassindex [BMI];
?>
```

Then the result of this code will be

- A. your weight is weight                      your weight is weight   your height is height   your BMI is BMI  
your height is height  
your BMI is BMI
- B. your weight is 65kg                      your weight is 65kg   your height is 1.8cm   your BMI is BMI 20.3  
your height is 1.8cm  
your BMI is BMI 20.3

21. a function in PHP are
- A. used once in a program
  - B. **built-in as well as user defined**
  - C. run automatically when page loaded
  - D. all
22. in PHP session, session\_unset(); will used to;
- A. create session
  - B. destroy all the session
  - C. **remove all session**
  - D. start session
23. from the following which one is used to call value within an object
- A. \$this->new ();
  - B. **\$this**
  - C. new class\_name();
  - D. class\_name ->object\_name();
24. a representation of data in which the implementation details are hidden is called
- A. encapsulation
  - B. **abstraction**
  - C. polymorphism
  - D. inheritance
25. the preg\_match PHP built-in function will be used to
- A. perform a pattern match on string followed by the splitting of result
  - B. perform a pattern match on string followed by replacing the match with specified text
  - C. **execute a pattern match on string**
  - D. display content from database when it matches with input field
26. It is referring to a concept where we encapsulate all the data and member functions together to form an object
- A. **encapsulation**
  - B. abstraction
  - C. polymorphism
  - D. inheritance
27. from the following PHP statement, which one is takes all the contents in the specified file and copies it into the file that uses the statement.
- A. include
  - B. require
  - C. require\_once
  - D. **all**

28. which one of the following used PHP form to upload different types files

- A. METHOD
- C. ENCTYPE
- B. POST
- D. ACTION

29. an attribute specifies the path where the form is to be submitted

- A. METHOD
- C. ENCTYPE
- B. POST
- D. ACTION

30. \_\_\_\_ is specifies the directory where the file is going to be placed during we upload files in PHP form.

- A. \$target\_file
- B. \$target\_dir
- C. \$basename
- D. \$\_FILES

